

Tutorial-1(4C)

	Random Process	
1.	Random variable X has a Gaussian PDF with mean and variance (3, 4), and Random variable Y also has a Gaussian PDF with mean and variance of (-1, 2). If X and Y are independent, determine the mean of the two random variables $Z = X - Y$ and $W = 2X + 3Y$.	2
2.	Let $X(t) = 8 \cos(2\pi t + \theta)$, where θ is a uniformly distributed in the interval $[0, 2\pi]$. The total power of the process is _____ W.	2
3.	A random process has a sample function of the form $X(t) = \begin{cases} A, & 0 \leq t \leq 1 \\ 0, & \text{Elsewhere} \end{cases}$ Where A is a random variable that is uniformly distributed from 0 to 10. The autocorrelation of the process is _____.	2
4.	Discuss the properties of autocorrelation function and power spectral density.	2
5.	The random process $X(t) = A \sin(\omega_c t + \Theta)$. The amplitude A and the frequency ω_c are constants and phase Θ is a random variable uniformly distributed over $[0, 2\pi]$. Check if X(t) is ergodic in both mean and autocorrelation.	5
6.	A modulated random signal $Y(t) = A X(t) \cos(\omega_c t + \Theta)$. Where X(t) is the random message signal and is a zero-mean stationary random process. The carrier amplitude A and the frequency ω_c are constants and phase Θ is a random variable uniformly distributed over $[-\pi, \pi]$. Assuming that X(t) and Θ are independent, find the mean, autocorrelation and power spectrum of Y(t). Check if Y(t) is a WSS random process.	5
7.	Derive the expression for Wiener Kinchine theorem in discrete form.	5
8.	Let X(t) and Y(t) are both zero mean and WSS random processes. Consider the random process $Z(t) = X(t) + Y(t)$. If X(t) and Y(t) are orthogonal, determine the power spectrum of Z(t) and the mean square value of Z(t).	5
9.	Let X(t) and Y(t) are defined by $X(t) = A \cos(\omega t) + B \sin(\omega t)$ and $Y(t) = B \cos(\omega t) - A \sin(\omega t)$. Where 'w' is constant and A and B are independent Random Variables both having zero mean and variance 2 . Find the cross correlation of X(t) and Y(t).	5
10	The input X(t) to the RC filter (Lowpass filter) is a white noise process. Determine i. The power spectrum of the output process Y(t), at the output of the filter. ii. Mean Square value of Y(t)	5
11	Discuss the properties of auto-covariance, function and power spectral density.	5
12	Write short notes on :WSS, SSS, Ergodicity, Ensemble	5